

BlackEnergy Lab[端點取證](Volatility2.6)

一家跨國公司遭受網路攻擊，導致敏感資料被盜。這次攻擊使用了先前未曾見過的 BlackEnergy v2 惡意軟體變種。該公司的安全團隊已從受感染的電腦獲取記憶體轉儲，並希望您作為 SOC 分析師能夠對轉儲進行專業分析，以了解此攻擊的範圍和影響。

Tool:

Volatility

問題 1

哪種波動率特徵最適合這台機器？

指令:volatility_2.6_win64_standalone.exe -f C:\Users\user\Downloads\temp_extract_dir\CYBERDEF-567078-20230213-171333.raw imageinfo

```
C:\Users\user\Desktop\鑑識工具\volatility_2.6_win64_standalone>volatility_2.6_win64_standalone.exe -f C:\Users\user\Downloads\temp_extract_dir\CYBERDEF-567078-20230213-171333.raw imageinfo
Volatility Foundation Volatility Framework 2.6
INFO : volatility.debug : Determining profile based on KDBG search...
Suggested Profile(s) : WinXPSP2x86, WinXPSP3x86 (Instantiated with WinXPSP2x86)
AS Layer1 : IA32PagedMemory (Kernel AS)
AS Layer2 : FileAddressSpace (C:\Users\user\Downloads\temp_extract_dir\CYBERDEF-567078-20230213-171333.raw)
PAE type : No PAE
DTB : 0x39000L
KDBG : 0x8054cde0L
Number of Processors : 1
Image Type (Service Pack) : 3
KPCR for CPU 0 : 0xffdf000L
KUSER_SHARED_DATA : 0xffdf0000L
Image date and time : 2023-02-13 18:29:11 UTC+0000
Image local date and time : 2023-02-13 10:29:11 -0800
```

WinXPSP2x86

第二季

取得影像時有多少個進程正在運作？

指令:volatility_2.6_win64_standalone.exe -f C:\Users\user\Downloads\temp_extract_dir\CYBERDEF-567078-20230213-171333.raw --profile=WinXPSP2x86 pslist

```
C:\Users\user\Desktop\鑑識工具\volatility_2.6_win64_standalone>volatility_2.6_win64_standalone.exe -f C:\Users\user\Downloads\temp_extract_dir\CYBERDEF-567078-20230213-171333.raw --profile=WinXPSP2x86 pslist
Volatility Foundation Volatility Framework 2.6
Offset(V) Name PID PPID Thds Hnds Sess Wow64 Start Exit
-----
0x89c037f8 System 4 0 55 245 ----- 0 0 2023-02-14 04:54:15 UTC+0000
0x89965020 smss.exe 368 4 3 19 ----- 0 0 2023-02-14 04:54:15 UTC+0000
0x89a98da0 csrss.exe 592 368 11 321 0 0 2023-02-14 04:54:15 UTC+0000
0x89a88da0 winlogon.exe 616 368 18 508 0 0 2023-02-14 04:54:15 UTC+0000
0x89938998 services.exe 660 616 15 240 0 0 2023-02-14 04:54:15 UTC+0000
0x89aa0020 lsass.exe 672 616 21 335 0 0 2023-02-14 04:54:15 UTC+0000
0x89aaa3d8 VBoxService.exe 832 660 9 115 0 0 2023-02-14 04:54:15 UTC+0000
0x89aab590 svchost.exe 880 660 21 295 0 0 2023-02-13 17:54:16 UTC+0000
0x89a9f6f8 svchost.exe 968 660 10 244 0 0 2023-02-13 17:54:17 UTC+0000
0x89730da0 svchost.exe 1060 660 51 1072 0 0 2023-02-13 17:54:17 UTC+0000
0x897289a8 svchost.exe 1108 660 5 78 0 0 2023-02-13 17:54:17 UTC+0000
0x899adda0 svchost.exe 1156 660 13 192 0 0 2023-02-13 17:54:17 UTC+0000
0x89733938 explorer.exe 1484 1440 14 489 0 0 2023-02-13 17:54:18 UTC+0000
0x897075d0 spoolsv.exe 1608 660 10 106 0 0 2023-02-13 17:54:18 UTC+0000
0x89694388 wscntfy.exe 480 1060 1 28 0 0 2023-02-13 17:54:30 UTC+0000
0x8969d2a0 alg.exe 540 660 5 102 0 0 2023-02-13 17:54:30 UTC+0000
0x89982da0 VBoxTray.exe 376 1484 13 125 0 0 2023-02-13 17:54:30 UTC+0000
0x8994a020 msmgs.exe 636 1484 2 157 0 0 2023-02-13 17:54:30 UTC+0000
0x89a0b2f0 taskmgr.exe 1880 1484 0 ----- 0 0 2023-02-13 18:25:15 UTC+0000
0x899dd740 rootkit.exe 964 1484 0 ----- 0 0 2023-02-13 18:25:26 UTC+0000
0x89a18da0 cmd.exe 1960 964 0 ----- 0 0 2023-02-13 18:25:26 UTC+0000
0x896c5020 notepad.exe 528 1484 0 ----- 0 0 2023-02-13 18:26:55 UTC+0000
0x89a0d180 notepad.exe 1432 1484 0 ----- 0 0 2023-02-13 18:28:25 UTC+0000
0x899e6da0 notepad.exe 1444 1484 0 ----- 0 0 2023-02-13 18:28:42 UTC+0000
0x89a0fda0 DumpIt.exe 276 1484 1 25 0 0 2023-02-13 18:29:08 UTC+0000
```

扣掉exit的部分。

19

第三季

的進程 ID 是什麼cmd.exe？

同上

1960

第四季

最可疑的進程名稱是？

同Q2

rootkit.exe

問5

哪個進程最有可能出現程式碼注入？

指令:volatility_2.6_win64_standalone.exe -f C:\Users\user\Downloads\temp_extract_dir\CYBERDEF-567078-20230213-171333.raw --profile=WinXPSP2x86 malfind

```
Process: svchost.exe Pid: 880 Address: 0x980000
Vad Tag: VadS Protection: PAGE_EXECUTE_READWRITE
Flags: CommitCharge: 9, MemCommit: 1, PrivateMemory: 1, Protection: 6

0x00980000  4d 5a 90 00 03 00 00 00 04 00 00 00 ff ff 00 00  MZ. ....
0x00980010  b8 00 00 00 00 00 00 00 40 00 00 00 00 00 00 00  .....@.....
0x00980020  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  .....
0x00980030  00 00 00 00 00 00 00 00 00 00 00 00 f8 00 00 00  .....

0x00980000  4d          DEC EBP
0x00980001  5a          POP EDX
0x00980002  90          NOP
0x00980003  0003        ADD [EBX], AL
0x00980005  0000        ADD [EAX], AL
0x00980007  000400      ADD [EAX+EAX], AL
0x0098000a  0000        ADD [EAX], AL
0x0098000c  ff          DB 0xff
0x0098000d  ff00        INC DWORD [EAX]
0x0098000f  00b800000000 ADD [EAX+0x0], BH
0x00980015  0000        ADD [EAX], AL
0x00980017  004000      ADD [EAX+0x0], AL
```

Process: svchost.exe Pid: 880 Address: 0x980000

svchost.exe

問6

在最近的進程中引用了一個奇怪的檔案。請提供該文件的完整路徑。

指令:volatility_2.6_win64_standalone.exe -f C:\Users\user\Downloads\temp_extract_dir\CYBERDEF-567078-20230213-171333.raw --profile=WinXPSP2x86 handles -p 880 -t file

```
C:\Users\user\Desktop\鑑識工具\volatility_2.6_win64_standalone>volatility_2.6_win64_standalone.exe -f C:\Users\user\Downloads\temp_extract_dir\CYBERDEF-567078-20230213-171333.raw --profile=WinXPSP2x86 handles -p 880 -t file
Volatility Foundation Volatility Framework 2.6
Offset(V)      Pid      Handle      Access Type      Details
-----
0x89a28890     880      0xc      0x100020 File      \Device\HarddiskVolume1\WINDOWS\system32
0x89a1a6f8     880      0x50      0x100001 File      \Device\KsecDD
0x89937358     880      0x68      0x100020 File      \Device\HarddiskVolume1\WINDOWS\WinSxS\x86_Microsoft.Windows.Common-Controls_6595b64144ccf1df_6.0.26
00_5512_x-ww_35d4ce83
0x899d0250     880      0xbc      0x12019f File      \Device\NamedPipe\nt\NtControlPipe2
0x89a17a50     880      0x100      0x100000 File      \Device\Dfs
0x89732cb8     880      0x158      0x12019f File      \Device\NamedPipe\lsarpc
0x8969fee0     880      0x274      0x12019f File      \Device\Termdd
0x89ab3478     880      0x294      0x12019f File      \Device\Termdd
0x89ab3978     880      0x29c      0x12019f File      \Device\Termdd
0x896bcb18     880      0x2b8      0x12019f File      \Device\NamedPipe\Ctx_WinStation_API_service
0x8997a248     880      0x2bc      0x12019f File      \Device\NamedPipe\Ctx_WinStation_API_service
0x899a24b0     880      0x304      0x12019f File      \Device\Termdd
0x89a00f90     880      0x33e      0x12019f File      \Device\{9DD6AFA1-8646-4720-836B-EDCB1085864A}
0x89af0cf0     880      0x340      0x12019f File      \Device\HarddiskVolume1\WINDOWS\system32\drivers\str.sys
0x89993f90     880      0x3d8      0x100020 File      \Device\HarddiskVolume1\WINDOWS\WinSxS\x86_Microsoft.Windows.Common-Controls_6595b64144ccf1df_6.0.26
00_5512_x-ww_35d4ce83
0x89958b78     880      0x3e4      0x12019f File      \Device\HarddiskVolume1\WINDOWS\system32\config\systemprofile\Local Settings\Temporary Internet File
s\Content.IE5\index.dat
0x899fe2e0     880      0x3f8      0x12019f File      \Device\HarddiskVolume1\WINDOWS\system32\config\systemprofile\Cookies\index.dat
0x89a492e8     880      0x400      0x12019f File      \Device\HarddiskVolume1\WINDOWS\system32\config\systemprofile\Local Settings\History\History.IE5\ind
ex.dat
0x896811d8     880      0x424      0x100020 File      \Device\HarddiskVolume1\WINDOWS\WinSxS\x86_Microsoft.Windows.Common-Controls_6595b64144ccf1df_6.0.26
00_5512_x-ww_35d4ce83
0x89bb028      880      0x488      0x100020 File      \Device\HarddiskVolume1\WINDOWS\WinSxS\x86_Microsoft.Windows.Common-Controls_6595b64144ccf1df_6.0.26
00_5512_x-ww_35d4ce83
0x89999980     880      0x4a8      0x1200a0 File      \Device\NetBT_Tcpip_{B35F0A5F-EBC3-4B5D-800D-7C1B64B30F14}
```

C:\WINDOWS\system32\drivers\str.sys

問7

從最近的進程載入的注入的 DLL 檔案的名稱是什麼？

指令:volatility_2.6_win64_standalone.exe -f C:\Users\user\Downloads\temp_extract_dir\CYBERDEF-567078-20230213-171333.raw --profile=WinXPSP2x86 ldrmodules

找pid 880且 InLoad、InInit、InMem <=這3項為False

880	svchost.exe	0x77120000	True	True	True	\WINDOWS\system32\Greadt32.dll
880	svchost.exe	0x5cb70000	True	True	True	\WINDOWS\system32\shimeng.dll
880	svchost.exe	0x74980000	True	True	True	\WINDOWS\system32\msxml3.dll
880	svchost.exe	0x009a0000	False	False	False	\WINDOWS\system32\msxml3r.dll
880	svchost.exe	0x77e70000	True	True	True	\WINDOWS\system32\rpcrt4.dll
880	svchost.exe	0x769c0000	True	True	True	\WINDOWS\system32\userenv.dll
880	svchost.exe	0x7c800000	True	True	True	\WINDOWS\system32\kernel32.dll
880	svchost.exe	0x76fd0000	True	True	True	\WINDOWS\system32\clbcatq.dll

msxml3r.dll

問8

注入的DLL的基底位址是多少？

問Q5

0x980000